

Robust Streaming ASR with Decoupled Separation and Recognition

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Abstract

Real-time speech transcription that is robust to background noise, room reverberation, and interfering speakers is critical for many real-world automatic speech recognition (ASR) applications. To address these interferences, multi-condition training (MCT) is commonly employed for training an ASR model, but it requires large-scale training data and degrades recognition performance on clean speech. In this work, we propose a decoupled framework that leverages an online speech separation frontend, and a streaming ASR backend trained only on clean speech, for robust streaming ASR. We benchmark the proposed approach on LibriSpeech, CHiME-4, and LibriCSS datasets, and demonstrate that the decoupled framework consistently outperforms the MCT-based baselines for streaming ASR. Our approach is also shown to generalize to large pre-trained speech models, offering a modular approach for robust streaming ASR that avoids task-specific re-training and MCT.

Index Terms: CHiME-4, LibriCSS, online speech separation, robust ASR, streaming ASR

1. Introduction

Over the past decades, automatic speech recognition (ASR) has evolved from Gaussian mixture model (GMM) and hidden Markov model (HMM) systems [1, 2] to deep learning based end-to-end architectures [3, 4], leading to substantial improvement in recognition accuracy and reaching human-level performance in relatively clean conditions [5]. In modern speech applications, streaming ASR has become increasingly needed for voice assistants, real-time captioning, and spoken dialogue systems, where low latency and reliable transcription are both important [6].

The transition toward streaming ASR has been marked by a series of architectural advancements, including recurrent neural network transducer (RNN-T) [7] and chunk-based attention mechanisms [8]. The evaluation of streaming ASR systems is mainly focused on the word error rate (WER) and algorithmic latency [9]. However, real-world deployment is subject to background noise, room reverberation, and interfering speakers, and streaming ASR in such acoustic conditions is underexplored with existing systems. While general ASR robustness has been examined through frameworks such as Speech Robust Bench [10], a benchmark for evaluating the robustness of streaming ASR remains lacking. This gap can be problematic as streaming models, constrained by limited future context, often exhibit different failure modes under noise compared with offline models. Thus, we investigate streaming ASR systems beyond clean-speech datasets to evaluate their robustness.

To mitigate the impact of acoustic interference, the most prevalent approach is multi-condition training (MCT), where an

ASR model is exposed to a large, diverse amount of background noise during training. While MCT improves generalization, it often fails to reach the performance of clean-trained models when tested on clean speech. An alternative approach involves integrating a speech separation (SS) frontend. However, the SS frontend and ASR backend are subject to a mismatch effect, where the separated speech deviates from the ASR training data. In recent years, as SS progresses, prior studies [11, 12] find that decoupled SS and ASR trained only on clean speech can outperform MCT-based systems.

In this paper, we present a decoupled framework for robust streaming ASR, consisting of an online SS frontend and a streaming ASR backend trained only on clean speech. We evaluate the system comprehensively and benchmark the results on both simulated and real data. Specifically, we investigate online SS frontends such as open-sourced DPDFNet [13] and TF-CrossNet [14]. For the ASR backend, we propose FastMambaformer, a novel architecture built upon FastConformer [15, 16] and leveraging Mamba [17] to improve the modeling capability in a Conformer [18] encoder. In addition, we employ multiple representative pre-trained streaming ASR models, such as NeMo [19] pre-trained FastConformer [15, 16]. We train all models on LibriSpeech [20] and test additionally on CHiME-4 [21] and LibriCSS [22] datasets, demonstrating the cross-corpus generalizability of the decoupled framework. We show that, given a sufficiently strong online SS frontend, we can eliminate MCT with a much smaller cost of ASR training, and much more flexibility.

The main contributions are summarized as follows. First, we establish a benchmark for robust streaming ASR by comprehensively evaluating multiple online SS and streaming ASR models under streaming constraints (zero lookahead). Second, we propose FastMambaformer, a new streaming ASR model that outperforms FastConformer. Third, we demonstrate that a decoupled system, utilizing a strong online SS frontend and clean-trained ASR backend, is able to outperform MCT-based models in streaming conditions, which has not been reported in prior studies. Fourth, we show that the benefit of the decoupled framework is architecture-agnostic by integrating different frontend-backend model architectures. This suggests a promising direction for improving robustness without task-specific re-training or fine-tuning, and the potential to benefit audio-based foundation models with an SS frontend.

The remainder of the paper is organized as follows. Section 2 describes the architectures of the frontends and backends. Section 3 describes the experimental setup and implementation details. Section 4 presents the results and comparisons. Conclusions and discussions are made in Section 5.

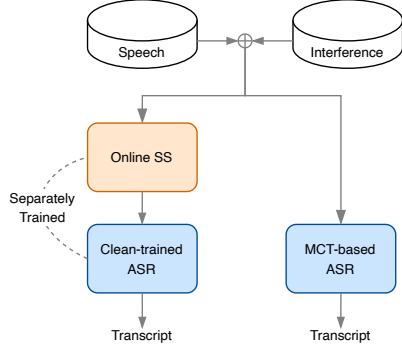


Figure 1: Comparison between decoupled system (left branch) and MCT-based system (right branch).

2. Methods

2.1. Overall Decoupled Framework

The comparison between the decoupled framework and the MCT-based system is shown in Figure 1. The online SS model and MCT-based ASR model are trained on a comparable amount of data. The MCT-based model mitigates the interference but achieves suboptimal performance on clean speech. In the decoupled framework, the online SS frontend and streaming ASR backends are trained separately, with full flexibility for improving either model when a more advanced model is available, without additional task-specific re-training or fine-tuning.

2.2. Online Speech Separation Frontend

2.2.1. DPDFNet

We employ DPDFNet [13], an online SS model that extends the DeepFilterNet2 architecture [23] with dual-path blocks in the encoder. It strengthens long-range temporal and cross-band modeling while preserving the original framework compared with DeepFilterNet2. The model uses grouped temporal convolutions and dual-path recurrent units, merges high-frequency bands through an equivalent rectangular bandwidth filter bank, and applies temporal recurrent attention.

2.2.2. TF-CrossNet

We utilize TF-CrossNet [14] as another SS frontend. TF-CrossNet consists of a cross-temporal module, a cross-band module, and positional encoding. It performs complex spectral mapping by predicting the real and imaginary (RI) parts of the short-time Fourier transform (STFT) of clean speech from the stacked RI parts of the STFT of noisy speech [24, 25]. The separated waveform is generated by performing an inverse STFT on the separated RI parts. The TF-CrossNet is designed to perform offline SS. By adding causal attention masks and making all convolution layers causal, TF-CrossNet is modified into an online SS model with zero frame lookahead.

2.3. Streaming ASR Backend

2.3.1. FastMambaformer

We propose FastMambaformer, a new ASR architecture based on the FastConformer [15, 16] backbone, which is a state-of-the-art open-sourced streaming ASR model. We replace the convolutional module with Mamba [17] to leverage better se-

quence information than the convolutional module, in addition to the multi-head attention mechanism. Mamba is a selective state-space model (SSM), which is designed to enable efficient, linear-time sequence modeling by parameterizing the state-space dynamics as functions of the input at each time step. Unlike traditional time-invariant SSMs, where state transition and output matrices remain fixed, Mamba’s parameters are dynamically generated from the current input, allowing the model to selectively propagate or attenuate relevant information across long contexts. This gives better sequence modeling capability than the convolutional module in FastConformer.

2.3.2. SimulStreaming

SimulStreaming [26] is a streaming ASR model built on Whisper [5] and combines Simul-Whisper [27] and Whisper-Streaming [28], leveraging the knowledge from the Whisper large-v3 model but adapting it for streaming ASR, which is the state-of-the-art Whisper variant for streaming ASR. The adaptation is done by segmenting the input speech using a buffering mechanism and splitting the continuous audio into short, incremental chunks that are appended into a rolling audio buffer. At each update step, the accumulated buffer is encoded, and the decoder generates output tokens conditioned on the partial audio and previously decoded text, following a simultaneous decoding policy to decide how much of the decoded hypothesis can be emitted without waiting for future frames. The model maintains multiple internal buffers to handle overlapping contexts across chunks and to preserve previously decoded text for continuity.

3. Experimental Setup

3.1. Datasets

To evaluate the robust streaming ASR systems, we experiment on the LibriSpeech [20], CHiME-4 [21], and LibriCSS [22] datasets. All models are trained only on LibriSpeech and evaluated on three datasets. This provides insights into the cross-corpus generalizability and the robustness to unseen background noise, room reverberation, and overlapping speakers for each system. The sampling rate for all utterances is 16 kHz.

For LibriSpeech, the training data is generated by mixing all 960 hr training sets with noise segments randomly sampled from the 10k non-speech sounds of a sound effect library (<https://sound-ideas.com>) with signal-to-noise ratio (SNR) uniformly selected in $[-5, 0]$ and $[0, 10]$ dB ranges, with 50% probability for each range. Validation data is generated by mixing all development datasets with the factory noise from the NOISEX-92 dataset [29] at -5 dB SNR. The evaluation sets are generated by mixing the test-other set with ADTBabble and ADTCafeteria noises (<https://auditec.com>) at $\{-5, -2, 0, 2, 5, 10\}$ dB SNR levels.

We evaluate on the CHiME-4 dataset, which comprises simulated and real noisy speech using the Wall Street Journal (WSJ) [30] text material. The test set consists of 1320 simulated and 1320 real noisy utterances from 4 different speakers. Real data is recorded in noisy environments (bus, cafe, pedestrian area, and street) spoken by real talkers. Simulated data is generated by mixing recorded speech via a close-talking microphone and separately recorded noises. We select the official 1-ch track data for evaluation.

For the LibriCSS dataset, we select utterance-wise evaluation, which is more suitable for monaural streaming ASR. From the original 7-ch recordings, we extract the first channel follow-

Table 1: ASR (%WER) results comparison of different systems on LibriSpeech.

SS Frontend	ASR Backend	SNR (dB)						Avg
		-5	-2	0	2	5	10	
-	Clean-trained	99.8	96.2	90.4	80.5	61.4	32.7	76.8
-	Noisy-trained	67.3	48.9	38.3	30.1	21.7	15.4	36.9
-	NeMo Pre-trained	91.0	72.1	56.8	43.0	27.5	14.9	50.9
-	SimulStreaming	92.0	65.5	47.2	34.7	22.2	13.4	45.8
DPDFNet	Clean-trained	83.4	66.0	53.4	41.7	28.7	17.6	48.5
oTF-CrossNet		66.1	48.0	37.6	29.4	21.1	14.4	36.1
TF-CrossNet		54.4	38.0	29.6	23.6	17.7	12.7	29.3
DPDFNet	NeMo Pre-trained	75.9	56.7	44.2	33.5	22.7	13.3	41.0
oTF-CrossNet		60.4	41.8	31.6	24.1	16.7	10.7	30.9
TF-CrossNet		50.0	33.4	25.3	19.7	14.2	9.7	25.4
DPDFNet	SimulStreaming	83.3	62.6	47.9	35.3	23.5	14.4	44.5
oTF-CrossNet		67.5	45.7	33.2	25.3	17.3	11.9	33.5
TF-CrossNet		53.5	34.8	25.8	19.9	14.7	10.9	26.6

ing the official script¹. We evaluate all utterances in split of OS (no overlap and inter-utterance silence ranges in [0.1, 0.5] s), 0L (no overlap and inter-utterance silence ranges in [2.9, 3.0] s), and speaker overlap ratio at 10%, 20%, 30%, and 40%.

3.2. Implementation Details

3.2.1. Frontend Configuration

For DPDFNet, we utilize the pre-trained dpdfnet8 model² with 3.54M parameters for the best quality, following the official script.

For TF-CrossNet, we train two models, one online with zero lookahead, and one offline with full-context, denoted as **oTF-CrossNet** and **TF-CrossNet**, respectively. Both models are trained on noisy LibriSpeech, and the training data is generated dynamically for every epoch. During training, a 4-second segment is randomly selected for all utterances. We use 12 TF-CrossNet blocks for each model, with hidden size set to 192, squeeze dimension set to 16, convolutional feedforward dimension set to 384, and 4 self-attention heads. Two models have 7.95M parameters and are both trained for 50 epochs with a batch size of 32. We utilize the Adam optimizer with a maximum learning rate of $2e^{-4}$. The learning rate starts from $2e^{-4}$ for 16 epochs, and then gradually decays to $2e^{-5}$. Models are trained on 8 NVIDIA H100 GPUs.

3.2.2. Backend Configuration

For FastMambaformer, we follow the same configuration with [16] except integrating a Mamba block in each Conformer block to replace the convolutional module. For each Mamba block, the SSM state expansion factor is set to 16, the local convolution width to 4, and the block expansion factor to 2. We train two FastMambaformer models, one on clean speech and the other with MCT, denoted as **clean-trained** and **noisy-trained**, respectively. Both models have 130M parameters. The clean-trained model is trained on clean LibriSpeech only. The noisy-trained model serves as the baseline with MCT on both clean and noisy speech. To make the baseline training data comparable with the decoupled system, the noisy-trained ASR model sees the same duration of data as the online SS frontend and the clean-trained ASR backend training data. We use the same configuration as [16], and use the Nvidia NeMo

toolkit [19] to train each model for 600 epochs on 8 NVIDIA H100 GPUs. The model utilizes multiple lookaheads during training, but zero lookahead during inference. All FastMambaformer models use the RNN-T decoder in this study.

For pre-trained models, we utilize the open-sourced streaming FastConformer with multiple lookaheads from Nvidia NeMo (HuggingFace model name `stt_en_fastconformer_hybrid_large_streaming_multi`), denoted as **NeMo pre-trained**. The NeMo pre-trained model has 114M parameters and is trained on NeMo ASRNET, comprising several thousand hours of English speech. During inference, the lookahead is set to 0 ms, and the decoder is RNN-T. We also utilize **SimulStreaming**³ as another streaming ASR backend, leveraging the knowledge from the Whisper foundation ASR model [5], but it is capable of streaming ASR. We follow the default configuration of SimulStreaming for transcription, and it has 1.5B parameters.

4. Results and Discussions

4.1. Results on LibriSpeech

First, we compare the results on the LibriSpeech test-other set. In [16], FastConformer trained on LibriSpeech achieves 10.6%/9.5% WER with 0 ms lookahead with connectionist temporal classification (CTC)/RNN-T decoder. Clean-trained FastMambaformer outperforms it with 10.3%/9.1% WER, demonstrating the effectiveness of leveraging Mamba. Noisy-trained model achieves 12.5%/10.9% WER, higher than the clean-trained model, showing that MCT degrades performance on clean speech.

For SS metrics, the average short-time objective intelligibility (STOI) [31] and perceptual evaluation of speech quality (PESQ) [32] improvement of TF-CrossNet is 21% and 1.26. For online models, oTF-CrossNet and DPDFNet achieve 18% and 12% improvement in STOI, and 0.89 and 0.48 improvement in PESQ, respectively. Table 1 presents the average WER comparison between different systems on ADTBabble and ADTCafeteria noises. Without a frontend, the NeMo pre-trained model and SimulStreaming all outperform the clean-trained model due to their larger training set. The noisy-trained baseline outperforms all other models significantly with a 36.9% WER, showing the effectiveness of MCT to mitigate background noise.

For the clean-trained backend, coupled with DPDFNet, the

¹https://github.com/chenzhuo1011/libri_css

²<https://github.com/ceva-ip/DPDFNet>

³<https://github.com/ufal/SimulStreaming>

Table 3: ASR (%WER) results for different systems on LibriCSS utterance-wise evaluation (1-ch).

SS Frontend	ASR Backend	Overlap Ratio						Avg
		0S	0L	10%	20%	30%	40%	
-	Clean-trained	16.27	15.61	20.61	28.26	35.32	42.96	26.51
-	Noisy-trained	16.89	16.89	20.91	27.49	33.41	39.70	25.88
-	NeMo pre-trained	37.42	33.66	40.22	49.68	50.64	58.64	45.04
-	SimulStreaming	15.77	16.27	19.35	23.52	29.29	34.13	23.06
DPDFNet	Clean-trained	22.06	20.77	26.38	33.86	39.62	46.77	31.58
oTF-CrossNet		11.67	12.71	18.22	26.09	31.15	37.73	22.93
TF-CrossNet		7.75	8.25	12.24	17.93	21.84	25.39	15.57
DPDFNet	NeMo pre-trained	20.79	19.31	26.13	33.65	39.54	47.22	31.11
oTF-CrossNet		9.35	9.86	15.68	22.83	28.23	34.56	20.09
TF-CrossNet		6.09	5.84	9.49	14.60	18.58	22.44	12.84
DPDFNet	SimulStreaming	17.29	16.98	21.39	29.27	36.29	41.67	27.15
oTF-CrossNet		10.17	10.74	16.14	24.26	27.59	35.62	20.75
TF-CrossNet		6.87	7.25	9.76	15.89	19.23	22.74	13.62

average WER is 48.5%, which is better than the NeMo pre-trained model, but not as good as the noisy-trained baseline. However, when coupled with oTF-CrossNet, the 36.1% average WER is lower than the noisy-trained model, demonstrating that when the online SS frontend is sufficiently strong, a clean-trained backend can achieve better performance than a strong noisy-trained baseline with MCT. The finding also applies to the NeMo pre-trained model and SimulStreaming, when coupled with oTF-CrossNet as a frontend. When the TF-CrossNet is utilized as a frontend, all systems achieve even better performance because of the full-context benefit. The gap between the oTF-CrossNet and TF-CrossNet shows the difficulty of robust ASR in streaming conditions.

4.2. Results on CHiME-4

We load models trained on LibriSpeech and test directly on CHiME-4. This shows the cross-corpus generalizability and performance to unseen real noise and reverberation. Without a frontend, SimulStreaming outperforms all other ASR models with a 16.36% WER on the test real set. Trained on noisy LibriSpeech only, the noisy-trained baseline outperforms the NeMo pre-trained model.

With DPDFNet, the clean-trained model achieves 28.02% WER, outperforming the NeMo pre-trained model of 29.85% WER. With oTF-CrossNet, the system achieves 24.56% WER, even outperforming the noisy-trained baseline of 26.17% WER. This finding aligns with the results in Section 4.1. The decoupled framework provides full flexibility, and the modular-

ized system is much easier to update without task-specific re-training. For NeMo pre-trained and SimulStreaming backend, coupling them with oTF-CrossNet boosts the performance to 13.66% and 13.79% WER on the test real set, showing that for pre-trained large ASR models, coupling with an online SS can effectively improve the performance against noise and reverberation.

4.3. Results on LibriCSS

Similar to Section 4.2, we load models trained on LibriSpeech and evaluate on LibriCSS utterance-wise evaluation. Without frontend processing, SimulStreaming outperforms all other models because it has more training data. On LibriCSS, the clean-trained and noisy-trained models both outperform the NeMo pre-trained model, and the noisy-trained model has a lower WER on average. DPDFNet degrades the performance of all ASR backends, showing the outcome of the frontend processing artifact. However, with oTF-CrossNet, all models achieve a performance improvement in separating overlapping speakers. For the clean-trained backend, coupling with oTF-CrossNet yields a WER reduction from 26.51% to 22.93%. NeMo pre-trained model achieves 20.09% WER, significantly lower than the 45.04% WER without frontend. With oTF-CrossNet, SimulStreaming achieves 20.75% WER from 23.06% without frontend. The findings align with those in Section 4.2.

5. Concluding Remarks

In this work, we address robust streaming ASR with a decoupled framework where online SS frontend and streaming ASR backend are trained separately, with the streaming ASR model trained on clean speech only. We propose a new ASR model architecture, FastMambaformer, leveraging the strength of the FastConformer and Mamba. The proposed framework outperforms the baseline system trained with MCT. We demonstrate that the decoupled framework works on both simulated and real data, and benchmark the performance on CHiME-4 and LibriCSS datasets. The benchmark can be compared for online SS frontend performance, streaming ASR backend performance, or the entire robust streaming ASR system. The evaluation results on two open-sourced pre-trained ASR models show the potential of online SS as a frontend for modern audio language models.

Table 2: ASR (%WER) results comparison of different systems on CHiME-4 (1-ch Track).

SS Frontend	ASR Backend	Dev Set		Test Set	
		Simu	Real	Simu	Real
-	Clean-trained	43.05	38.36	51.13	57.35
-	Noisy-trained	23.06	20.20	26.27	26.17
-	NeMo Pre-trained	25.14	17.77	29.25	29.86
-	SimulStreaming	14.58	11.89	18.53	16.36
DPDFNet	Clean-trained	27.43	21.46	31.50	28.02
oTF-CrossNet		22.67	19.75	28.20	24.56
TF-CrossNet		19.24	16.81	23.47	20.69
DPDFNet	NeMo Pre-trained	14.04	10.45	19.05	15.68
oTF-CrossNet		11.76	9.65	16.98	13.66
TF-CrossNet		10.52	7.95	14.44	11.31
DPDFNet	SimulStreaming	16.28	11.87	19.94	15.42
oTF-CrossNet		14.07	11.46	17.87	13.79
TF-CrossNet		12.29	10.21	15.54	12.64

6. Acknowledgements

This research was supported in part by a National Science Foundation grant (ECCS-2125074), and the Ohio Supercomputer Center. All experiments were conducted at The Ohio State University.

7. Generative AI Use Disclosure

The authors utilized generative AI tools to polish the grammar and language of Section 1.

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